

Detecting Pneumonia in Chest X-Rays Using Deep Learning

A Machine Learning Capstone Project

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1. Introduction

Pneumonia is a severe respiratory infection and a leading cause of hospitalization and mortality worldwide, particularly among children under five and older adults. Diagnosis typically relies on radiologist interpretation of chest X-ray images, a process that can be time-consuming, subjective, and limited by the availability of trained specialists. In resource-constrained settings, delays in diagnosis can significantly worsen patient outcomes.

This project investigates whether deep learning models can accurately classify chest X-ray images as *Normal* or *Pneumonia*, with the goal of supporting clinicians through automated, reliable, and scalable diagnostic tools.

2. Problem Statement

The central question of this project is:

Can we develop a deep learning classifier that predicts whether a patient has pneumonia based on their chest X-ray image?

The problem is framed as a **binary image classification task**, where the model must distinguish between Normal chest X-rays and Pneumonia chest X-rays (bacterial or viral)

The primary objective is to minimize **false negatives**, as missing a pneumonia diagnosis poses significant clinical risk.

3. Data Description

The dataset used in this project is the Chest X-Ray Pneumonia dataset, originally sourced from Guangzhou Women and Children's Medical Center and made publicly available on Kaggle.

Dataset Characteristics

- **Total images:** ~5,000+
- **Classes:** Normal, Pneumonia
- **Format:** Grayscale JPEG images
- **Population:** Pediatric patients
- **Train/Test split:** Pre-defined by dataset authors
- **Class imbalance:** ~74% Pneumonia, ~26% Normal in the training set
- **Image variability:** Significant variation in resolution, brightness, and contrast

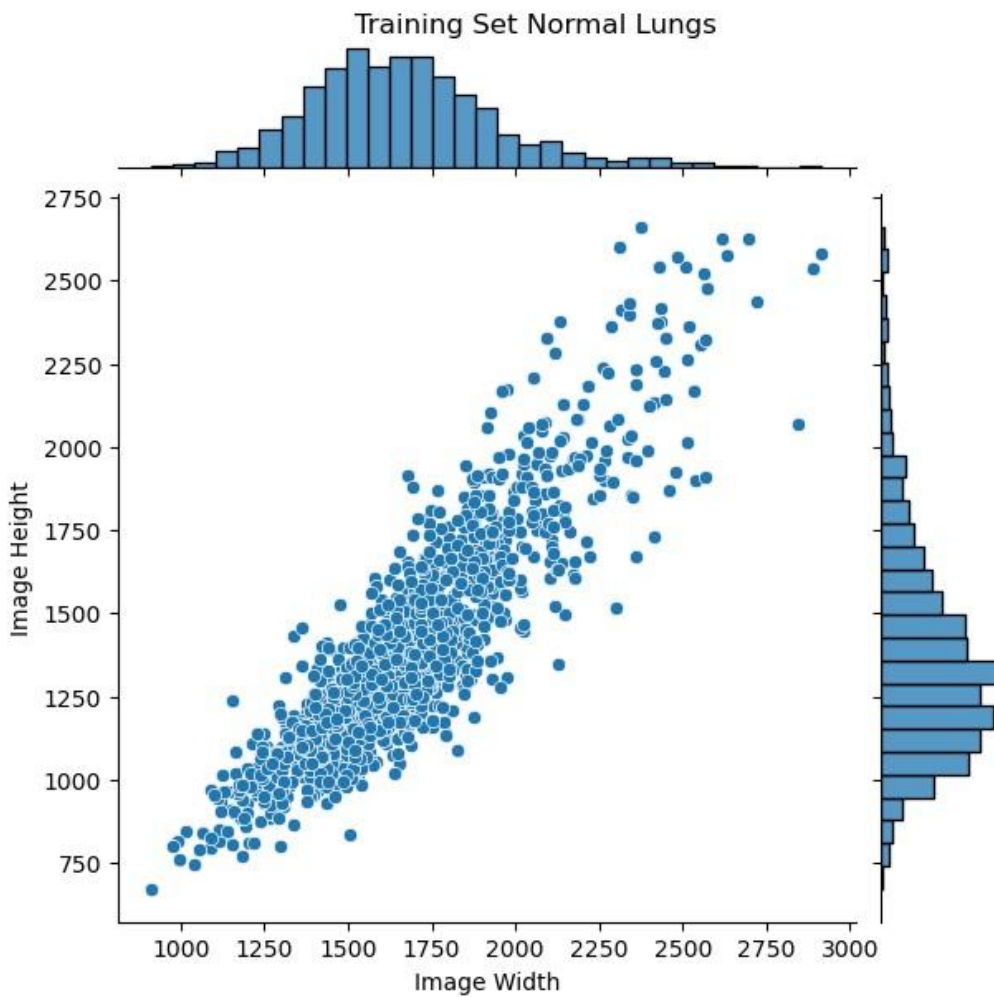
Implications

- Pediatric-only data limits generalization to adult populations
- Class imbalance requires careful evaluation of recall and precision
- Variability in image quality necessitates preprocessing and augmentation

4. Exploratory Data Analysis

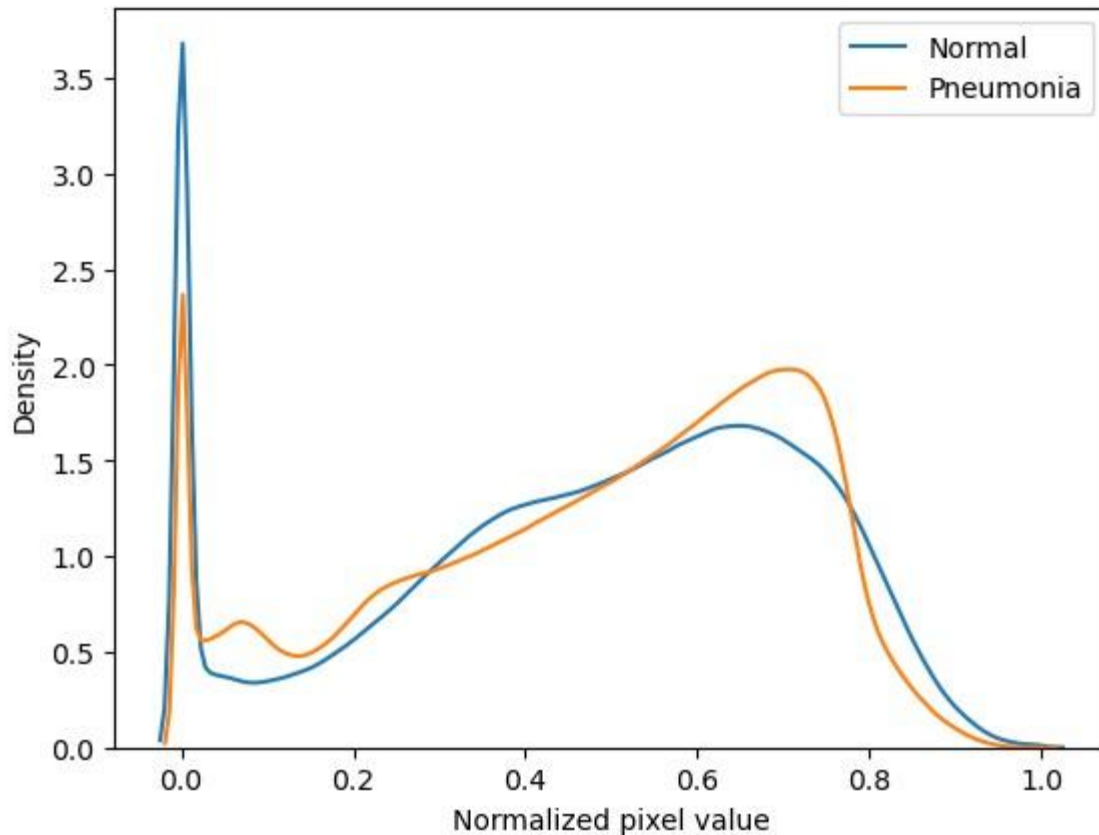
4.1 Image Resolution

Images vary widely in height and width, requiring resizing to a consistent input shape (128×128×1) for model training.



4.2 Pixel Intensity Distributions

- **Normal X-rays** tend to have more dark pixels (air-filled lungs).
- **Pneumonia X-rays** show more bright regions due to lung opacities.



These differences confirm that pixel-level patterns contain meaningful diagnostic signal.

5. Methodology

The modeling pipeline consists of:

5.1 Preprocessing

- Resize all images to **128×128**
- Convert to NumPy arrays
- Normalize pixel values
- Apply on-the-fly data augmentation:
 - Random horizontal flip
 - Random rotation ($\pm 2^\circ$)
 - Random zoom
 - Random contrast adjustments

Augmentation helps reduce overfitting and improves generalization.

5.2 Models Evaluated

(1) Multilayer Perceptron (MLP) Baseline

A simple, convolution-free neural network:

- Flattened input
- Two dense layers (512 units each)
- Dropout + L2 regularization
- Sigmoid output

Purpose: establish a baseline and test whether raw pixel intensities alone carry enough signal.

(2) Custom Convolutional Neural Network (CNN)

A standard CNN architecture with:

- Convolution + ReLU
- Max pooling
- Dense layers
- Sigmoid output

Purpose: leverage spatial feature extraction for improved performance.

(3) Transfer Learning Model: DenseNet121

A pretrained DenseNet121 model with:

- Frozen convolutional base
- Custom classification head
- Fine-tuning of top layers

Purpose: evaluate the power of transfer learning on medical imaging tasks.

5.3 Evaluation Metrics

Given the clinical context, the following metrics were prioritized:

- **Recall (Sensitivity):** ability to detect pneumonia cases
- **Precision:** reliability of positive predictions
- **Accuracy:** overall correctness
- **Confusion matrices:** error pattern analysis

Recall is the most important metric because false negatives are clinically dangerous.

6. Results

6.1 Model Comparison

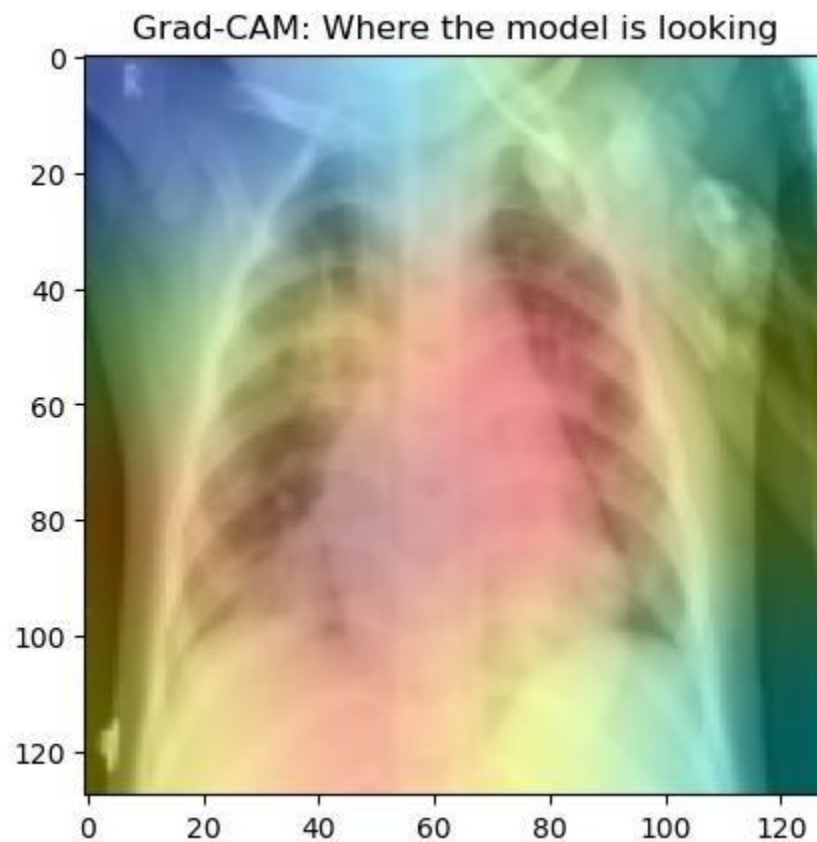
Model	Recall	Precision	Accuracy
MLP	96%	74%	76%
CNN	96%	85%	87%
DenseNet121	99%	83%	87%

Key Findings

- **DenseNet121 achieved the highest recall (99%),** minimizing false negatives.
- The CNN significantly improved precision over the MLP, confirming the value of convolutional feature extraction.
- Transfer learning provided the strongest overall performance, despite similar accuracy to the CNN.

7.1 Model Interpretability

Grad-CAM visualizations were used to understand where the DenseNet121 model focuses when predicting pneumonia.



- The model often highlights lung regions consistent with radiological signs of pneumonia.
- Some heatmaps show attention outside the lungs, suggesting potential reliance on spurious features (e.g., image borders, text markers).

This motivates future work in **lung segmentation** to constrain the model's attention to clinically relevant regions.

8. Discussion

8.1 Strengths

- High recall across all models
- Strong performance from transfer learning
- Effective use of augmentation to combat overfitting
- Grad-CAM interpretability analysis supports model trustworthiness

8.2 Limitations

- Dataset consists entirely of pediatric patients
- Single-institution data limits generalizability
- Class imbalance may inflate recall
- No external validation performed
- Model calibration not yet assessed

9. Recommendations

Use the DenseNet121 model as a high-recall screening tool

Because DenseNet121 achieved 99% recall, it is well-suited for initial triage. A client can deploy it to automatically flag X-rays that *may* show pneumonia so radiologists can prioritize those cases first.

Integrate the model into radiologist workflows as a decision-support aid

The model should not replace radiologists, but it can highlight suspicious regions (via Grad-CAM) and provide a second opinion.

Use the findings to guide future data collection and validation

Results show strong performance but limited generalizability due to pediatric, single-institution data. A client can begin collecting adult and multi-institution X-rays to perform external validation and strengthen the model for real-world deployment.

10. Future Work

10.1 External Validation

Test the model on an independent dataset from a different hospital or adult population to assess real-world generalization.

10.2 Lung Segmentation

Use a U-Net or similar model to isolate lung fields and reduce spurious correlations.

10.3 Larger and More Diverse Datasets

Incorporate multi-institution, multi-age datasets to improve robustness.

11. Conclusion

This project demonstrates that deep learning models — particularly transfer learning architectures like DenseNet121 — can effectively detect pneumonia in pediatric chest X-rays. DenseNet121 achieved **99% recall**, minimizing false negatives, which is the most critical metric in clinical diagnosis.

While the results are promising, real-world deployment requires additional steps, including external validation, and improved interpretability. Nonetheless, this work highlights the potential of deep learning to support radiologists and improve diagnostic workflows in healthcare.